
EXPANSIONS USING CALCULUS

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1 Introduction

We often use the expansion

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

or

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

However, do you know where they come from?

2 Polynomial Approximations

Let's try to approximate e^x as a polynomial. Say

$$e^x = \sum_{n=0}^{\infty} a_n x^n$$

Plugging in $x = 0$,

$$a_0 = 1$$

Now, we differentiate both sides to obtain

$$e^x = \sum_{n=0}^{\infty} n a_n x^{n-1}$$

Again, plugging in $x = 0$,

$$a_1 = 1$$

Repeating this process of differentiating and plugging in 0, we will get

$$a_n = \frac{1}{n!}$$

Try to prove this using induction.

Exercise 2.1. Find the Maclaurin series of $\sin x$ and $\cos x$ using their derivatives at 0.

3 Taylor series

Let us now try to prove a more general result.

Claim 1. The *Taylor series* about $x = a$ is

$$f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x)$$

where

$$R_n(x) = \frac{1}{n!} \int_a^x f^{(n+1)}(t)(x - t)^n dt$$

Proof. We proceed by induction on n .

For the base case, observe that

$$f(x) = f(a) + \int_a^x f'(t) dt$$

is just a consequence of the fundamental theorem of calculus.

Now, say our hypothesis holds for n . Observe that

$$R_n(x) = \frac{1}{n!} \int_a^x f^{(n+1)}(t)(x - t)^n dt$$

Set $u = f^{(n+1)}(t)$ and $v = -\frac{(x-t)^{n+1}}{n+1}$, so that

$$du = f^{(n+2)}(t) dt, \quad dv = (x - t)^n dt$$

Then

$$\int_a^x f^{(n+1)}(t)(x - t)^n dt = \left[-\frac{f^{(n+1)}(t)(x - t)^{n+1}}{n+1} \right]_a^x + \frac{1}{n+1} \int_a^x f^{(n+2)}(t)(x - t)^{n+1} dt$$

Therefore

$$R_n(x) = \frac{f^{(n+1)}(a)}{(n+1)!}(x - a)^{n+1} + \frac{1}{(n+1)!} \int_a^x f^{(n+2)}(t)(x - t)^{n+1} dt$$

Substituting back gives

$$f(x) = f(a) + f'(a)(x - a) + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + \frac{f^{(n+1)}(a)}{(n+1)!}(x - a)^{n+1} + \frac{1}{(n+1)!} \int_a^x f^{(n+2)}(t)(x - t)^{n+1} dt$$

As desired. □

Remark 1. For $a = 0$, the Taylor series is also called the *Maclaurin series*.

Exercise 3.1 (Challenge). Not every infinitely differentiable function is equal to its Taylor series.

Consider

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

It can be shown that $f^{(n)}(0) = 0$ for every $n \geq 0$.

What is the Maclaurin series of f ? Why does this show that Taylor series must be used carefully?

4 Further Reading

For better geometric intuition, check out 3blue1brown's video on Taylor series. A very helpful book is Tom Apostol's *Calculus, Volume 1*. Finally, for students who want a rigorous foundation in calculus and real analysis, Terence Tao's *Analysis I* is an excellent but demanding next step.